

Xuyue Zhang

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ECE Ph.D., Rice University | Computer Vision, Algorithmic and Intelligent Sensing | Neural Interface and Decoding

About Me

- 2nd-year PhD in computer vision, co-designing physical-inspired model and optic system for intelligent sensing and digital twin. Previously worked in brain-computer interface, focusing on brain signal decoding, bioelectrode recording and robotic exoskeleton system for patients. 2 published work in brain-computer interface, and one oral presentation in neural coding and control system. I'm determined to apply the edge-cutting world model and sensing knowledge to develop advanced technology in neuroengineering.
- **SKILLS (* indicating learned from course):**
 - **Algorithm:** Generative model for image, 3D splatting, deep learning, reinforcing learning*, implicit neural representation, algorithm for intelligent sensing and computer vision, algorithm for movement planning*, computational neuroscience*, statistical and multivariate analysis*
 - **Signal Process:** Fourier analysis, Fourier optics, deconvolution, single-neuron signal analysis (neural spikes), MEG/EEG/ECOG processing, neural coding
 - **Hardware:** Arduino coding, point spread function and aperture engineering, 4f optic system, physical sensing elements (IMU, piezo, LiDAR, event camera...), vision and imaging algorithms for robotics (Bug, SLAM...), embedded system*
 - **Biological & Neuroscience Experiments:** Neural interface implantation in rodent, brain signal recording, cell culture, immunostaining
- **Programming Language & Software:** MATLAB, Python (TensorFlow, PyTorch), C/C++
- **English Proficiency:** TOEFL (108)

EDUCATION & ACADEMIC EXPERIENCES

- **Rice University, Houston, TX** (Aug 2025 - Present)
 - Ph.D. in Electrical and Computer Engineering (GPA: 4.00/4.00)
 - Advisor: Ashok Veeraraghavan & Christopher Metzler (UMD)
- **Harvard University, Boston, MA** (Jul 2024 - Jun 2025)
 - Research fellow in School of Engineering and Applied Sciences
 - Advisor: Jia Liu
- **Tsinghua University, Beijing, China** (Aug 2020 - Jun 2024)
 - B.S. in Biological Sciences (GPA: 3.94/4.00, top 5%)
 - Advisor: Bo Hong (Department of Biomedical Engineering)
- **Harvard Medical School, Boston, MA** (Jun 2023 - Sep 2023)
 - Student intern in neurobiology department
 - Advisor: Bernardo Sabatini

RESEARCH EXPERIENCE

- **Rice Computational Imaging Lab** Department of Electrical and Computer Engineering, Rice University
Ph.D. Student (Aug 2025 - Present)
Field: Computer Vision, Intelligent Sensing, Physical Model
 - Project 1: Slit Aperture for 3D Scene Reconstruction**
 - Embedded multi-wiener deconvolution in a deep learning framework to reconstruct all-in-focus images from defocus and depth-variant scene, illustrate that slit aperture could strength reconstruction
 - The framework enables flat camera to reconstruct all-in-focus image with only 4 observations and single optical setting
 - Reconstruct and splatting 3D scene with defocused slit observations
 - Designed and built DMD-based tabletop prototype and validated it through real obscurants
 - Project 2: Implicit Neural Representation Network for Medical Imaging**

- Optimize a implicit neural network for cataract imaging and eliminating aberrations
 - Apply a neural representation to digitally achieve diffraction-limited imaging through strong spatial-variant aberrations
 - Build an optical system to optically correct these aberrations
- **Harvard Lab of Bioelectronics** School of Engineering and Applied Sciences, Harvard University
Research Fellow (Jul 2024 - Jun 2025)
Field: Flexible Brain-machine Interfaces, Neural Probes, Electrical Signal Processing and Analysis
Project: Long-term Monitoring of Treatment Processes Using a Robust and Flexible Brain-Machine Interface
 - Optimized the algorithms and constructed a database better for the long-term recording
 - Analyzed neuronal signals through the whole process of drug-therapy of epilepsy
 - Implanted neural interfaces in mice, induced their epilepsy by drug, and recorded their brain signals for months**Review: Material Selection and Device Design of Scalable Flexible Brain-Computer Interface**
 - Focused on material selection and innovative design for flexible brain-machine interfaces to optimize performance
 - Analyzed key electrical and physical parameters in neural probe structure and circuit design
 - **Tsinghua Laboratory of Brain and Intelligence** School of Biomedical Engineering, Tsinghua University
Undergraduate Research Intern (Sep 2022 - Jun 2024)
Field: Brain-Machine Interfaces, Cyborg Systems, Epileptic Network Analysis, Neurodynamic Modeling
Project 1: Clinical Brain-Machine Interface Experiments on Paraplegic Patients
 - For paraplegic patients, we implant brain-machine interface to record their ECoG data
 - We decoded their movement intention, then operated exoskeleton and mechanic arms to help them move, grab or hold items
 - We analyzed their clinical MEG and ECoG data, and found they have recovery after working with brain-machine interface and exoskeleton**Project 2: High-throughput Observation and Neurodynamics Modeling of In Vitro Epileptic Network**
 - Developed an in-vitro epilepsy system on multi-electrode array (MEA) and identify its key events
 - Verified that epileptic events resulted from alteration of network via neurodynamic modeling based on connectivity
 - **Bernardo Sabatini Lab** Department of Neurobiology, Harvard Medical School
Summer Intern Student (Jun 2023 - Oct 2023)
Field: Systems and Computational Neuroscience, Neural Coding, Control Systems
Project: An Closed-loop System for Investigating the Neural Coding Mechanisms of Termination in Mice
 - Built a real-time closed-loop system to detect the brain signal and executed neural modulation to mice based on live data
 - Trained machine learning models and found the most optimal in determining the termination with high accuracy

SELECTED PUBLICATIONS & CONFERENCE

- X. Lin, X. Zhang, J. Chen, J. Liu. "Material Selection and Device Design of Scalable Flexible Brain-Computer Interface: A Balance Between Electrical and Mechanical Performance." *Submitted to Advanced Materials, 2024*
- X.Lin, X. Zhang, et al. "Plastic-elastomer Heterostructure for Robust Flexible Brain-computer Interfaces." *bioRxiv, 2025*
- S. Liu, X. Zhang, B. Yang, B. Sabatini. "Searching for the Mechanism that Turns Oral Cooling into a Reinforcing Stimulus in Thirsty States." *Life Sciences, Basic Medicine, and Pharmaceutical Science Forum of Tsinghua University, 2023 (Oral Presentation)*

SELECTED COURSEWORK (4.0 for most listed)

- **Signal Processing & Engineering:** Signals and Systems, Microwave Engineering, Intelligent Mobile Systems, Physiological System Simulation and Modeling, Imaging and Vision for Robotics, Principle of Analogue Electronics
- **Algorithm:** Intro to Computer Vision, Computational Photography, Generative AI for Image Data, Pattern Recognition and Machine Learning, Algorithmic Robotics, Bioinformatics, Machine Learning, Data Structures and Algorithms
- **Mathematics:** Calculus, Linear Algebra, Probability and Statistics, Introduction to Causal Inference, Introduction to Bayesian Inference, Multivariate Statistical Analysis, Random Process and Application

- **Physics:** Imaging Optics, Physics for Scientists and Engineers, Physical Chemistry, Fourier Optics
- **Neuroscience & Neuroengineering:** Neural Modeling and Data Analysis, System and Computational Neuroscience, Biological Fundamental Neuroscience

SELECTED AWARDS & EXTRACURRICULAR ACTIVITIES

Awards & Honors:

- International Conference in Computational Photography (ICCP) Summer School Fellow *July 2026*
- Outstanding Member in Rice Chinese Students and Scholars Association (RCSSA) *hfill Nov 2025*
- National Scholarship (7 out of $\approx$ 400 eligible students) *Fall 2022*
- Overall Excellence Scholarship (Top 1 %) *Fall 2023*
- 28th China National Biology Olympiad, Silver medal (Top 0.5 % in China) *Aug 2019*

Leadership & Activities

- Member, IT Department of Rice Chinese Students and Scholars Association (RCSSA) *Sep 2025 – Present*
- Member, Tsinghua Xuetang Life Science Program (25 out of 90 students eligible) *Sep 2021 – Jun 2024*
- Minister and Team Leader, Student Volunteering Club of Cyrus Tang, Tsinghua University *May 2022 – Jun 2024*
- Member, Wispring Foundation (NGO focusing on education in poverty-stricken areas) *Sep 2023 – Present*